Software Requirements Specification

for

Trilingual Customer Support Ticket Management System

Version 1.1 approved

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Organization: [University / Department]
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Contents

Revision History

Name	Date	Change Reason	Version
Group 21	July 25, 2026	Revised to match the agreed project scope and page limit.	1.1

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines version 1.1 of the Trilingual Customer Support Ticket Management System. The system automates the initial handling of banking customer-support tickets by accepting text and an optional supporting image, predicting the ticket category, urgency, and sentiment, producing a context-aware draft response, and presenting all outputs to a support agent for review.

The term *trilingual* refers to English, Sinhala, and Tamil. The system must also handle Sinhala and Tamil written in Latin characters (Singlish and Tanglish) and messages that mix supported languages or scripts. Audio input and speech transcription are outside the approved scope.

1.2 Document Conventions

The word **shall** identifies a mandatory requirement, **should** a recommended requirement, and **may** an optional capability. Functional requirements use the prefix `FR`; nonfunctional requirements use `NFR`. Each requirement has a unique identifier and must be verifiable. Priorities are High, Medium, or Low.

1.3 Intended Audience and Reading Suggestions

This document is intended for the project supervisor, developers, data scientists, testers, UI/UX designers, support agents, and evaluators. Sections 1 and 2 describe scope and context; Section 3 defines external interfaces; Section 4 specifies system features; Sections 5 and 6 define quality, data, evaluation, and privacy requirements.

1.4 Product Scope

The product is a web-based research prototype and decision-support tool. Its objectives are to:

- accept English, Sinhala, Tamil, Singlish, Tanglish, and code-mixed ticket text;
- classify ticket intent using Banking77 categories or an approved subset;
- predict urgency/priority and customer sentiment using separately evaluated models;
- extract useful text or evidence from an optional image;
- generate an editable response draft in an appropriate supported language;
- route and track tickets while preserving human control over final decisions;
- compare model performance across language forms and prevent translation leakage.

The system shall not execute banking transactions, modify customer accounts, or send a model-generated response without authorised human review.

1.5 References

1. Karl E. Wiegers, *Software Requirements Specification Template*, 1999.
2. IEEE 830-1998, *IEEE Recommended Practice for Software Requirements Specifications.*
3. Banking77 dataset documentation and intent labels.
4. Group 21 project proposal, architecture, dataset plan, and model-evaluation plan.

2. Overall Description

2.1 Product Perspective

The product is a new academic prototype composed of a web interface, backend API, ticket database, multilingual text-processing pipeline, machine-learning inference services, optional OCR/vision processing, and an LLM-based response generator. It can operate as a standalone demonstration system; direct integration with a production bank or help-desk platform is not required for the initial release.

The principal flow is: submission → validation → multilingual preprocessing → category, priority, and sentiment prediction → optional image analysis → ticket routing → response drafting → agent review and resolution.

2.2 Product Functions

- authenticate authorised users and enforce simple roles;
- capture ticket text, metadata, and an optional image;
- preserve original text and create a model-ready representation;
- classify category, urgency, and sentiment;
- extract supplementary image evidence;
- create and route a ticket using model outputs and rules;
- generate, edit, approve, or reject a response draft;
- search and update ticket status;
- capture corrections and display evaluation results by language form.

2.3 User Classes and Characteristics

User Class	Characteristics and Responsibilities
Customer	Submits a multilingual ticket and optional image and views the ticket status or final response.
Support Agent	Reviews predictions and evidence, edits the response draft, changes priority/category when necessary, and resolves or escalates tickets.
Administrator / Project Team	Manages users, category mappings, model versions, confidence thresholds, evaluation datasets, and system configuration.
Project Evaluator	Uses authorised read-only views of system behaviour, metrics, and traceability evidence.

2.4 Operating Environment

The prototype shall support current desktop and mobile web browsers. The backend and ML services shall run on a Linux-compatible local server or cloud environment. Python-based model services, a REST API, a relational or document database, and optional Docker containers may be used. GPU resources may be used for training but shall not be mandatory for basic demonstration inference unless documented.

2.5 Design and Implementation Constraints

- Category-classification data shall originate from Banking77 and approved English, Sinhala, Tamil, Singlish, and Tanglish variants.
- All translations or transliterations derived from one Banking77 record shall remain in the same train, validation, or test split.
- Banking77 does not supply project-specific urgency and sentiment labels; these labels shall be obtained, annotated, or generated through a documented method before training those models.
- The system shall use Unicode-safe storage and multilingual tokenisation.
- Natural code-mixed data may be limited; synthetic mixing shall be distinguished from naturally written mixed-language evaluation data.
- Image input is optional and supplementary. Text processing shall continue if image analysis fails.
- LLM output shall be treated as a draft and shall not invent account or transaction facts.
- The solution shall remain feasible using the project team's available time, data, and computing resources.

2.6 User Documentation

The prototype shall provide a short customer submission guide, agent usage guide, administrator/model configuration guide, API or deployment notes, and a model-evaluation report.

2.7 Assumptions and Dependencies

The project assumes that translations preserve Banking77 intent, category mappings are approved, sufficient labelled data exists for each trained task, and users provide meaningful text. Dependencies may include a multilingual encoder, OCR/vision library, LLM service or local model, database, and object storage. Accuracy may decline for unseen slang, spelling variation, low-quality images, rare categories, or natural code-mixed patterns absent from training data.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide login, ticket submission, agent queue, ticket-detail, prediction-review, response-editing, status-management, evaluation, and basic administration views. The ticket-detail view shall show the original text, optional image, predicted category, urgency, sentiment, confidence values where available, extracted evidence, draft response, and agent correction controls. Sinhala and Tamil Unicode text shall render correctly.

3.2 Hardware Interfaces

Users shall provide text through a keyboard or touchscreen and may upload images from common cameras or storage devices. No specialised customer hardware is required. Server inference may use CPU or a compatible GPU.

3.3 Software Interfaces

The web client shall communicate with a backend API. The backend shall interface with the ticket database, multilingual inference services, optional OCR/vision service, and response-generation component. Interfaces shall use documented request/response schemas and include validation, model version, processing status, and meaningful error information. Production banking-core integration is outside the initial scope.

3.4 Communications Interfaces

Client-server and service-to-service communication shall use HTTPS where deployed over a network. Structured messages shall use UTF-8 JSON unless otherwise documented. Requests shall use timeouts and controlled error handling so that failure of an optional component does not lose the submitted ticket.

4. System Features

4.1 Authentication and Access Control

4.1.1 Description and Priority

Controls access to customer, agent, administration, and evaluation functions. **Priority:** Medium

4.1.2 Stimulus/Response Sequences

A user submits credentials; the system validates them and exposes the permitted functions. Invalid attempts return a generic error. An authorised administrator may create, deactivate, or change a role.

4.1.3 Functional Requirements

ID	Requirement
FR-AUTH-01	The system shall authenticate users before protected functions are available.
FR-AUTH-02	The system shall enforce server-side permissions for Customer, Agent, Administrator, and Evaluator roles.
FR-AUTH-03	The system shall support secure logout and session expiration.
FR-AUTH-04	The system shall store passwords using secure salted hashes and shall not log plaintext credentials.

4.2 Ticket Submission and Validation

4.2.1 Description and Priority

Captures multilingual ticket text, basic metadata, and an optional image. **Priority:** High

4.2.2 Stimulus/Response Sequences

The customer enters a description and optionally uploads an image. The system validates the input, stores the ticket, assigns an identifier, and starts analysis. Invalid content is rejected with corrective guidance.

4.2.3 Functional Requirements

ID	Requirement
FR-ING-01	The system shall accept English, Sinhala, Tamil, Singlish, Tanglish, and mixed-language ticket text.
FR-ING-02	The system shall preserve the original submitted text and store a separate model-ready representation.
FR-ING-03	The system shall accept an optional JPEG or PNG image subject to configured size and security limits.
FR-ING-04	The system shall reject empty, unsupported, corrupted, oversized, or unsafe input.
FR-ING-05	The system shall assign a unique ticket ID, submission time, and initial status.
FR-ING-06	The system shall not require audio input or speech transcription.

4.3 Multilingual Text Processing

4.3.1 Description and Priority

Creates consistent model input while retaining the linguistic information required for native, romanised, and mixed messages. **Priority:** High

4.3.2 Stimulus/Response Sequences

After validation, the system removes unsafe markup, normalises approved whitespace and Unicode variants, optionally identifies script/language spans, and tokenises the complete message using the configured multilingual pipeline.

4.3.3 Functional Requirements

ID	Requirement
FR-NLP-01	The system shall support Unicode English, Sinhala, and Tamil text.
FR-NLP-02	The system shall normalise only documented patterns without altering the intended meaning.
FR-NLP-03	The system shall process a code-mixed ticket as one message rather than requiring one sentence-level language label.
FR-NLP-04	The system shall process Singlish and Tanglish directly with the trained model or a documented transliteration/normalisation component.
FR-NLP-05	Language or script detection, when used, shall support preprocessing, analytics, or fallback and shall not be a mandatory gate before classification.
FR-NLP-06	The system shall flag extremely short, unreadable, or unsupported input for manual review.

4.4 Ticket Category Classification

4.4.1 Description and Priority

Predicts the Banking77 intent category or an approved mapped subset. **Priority:** High

4.4.2 Stimulus/Response Sequences

The model receives the processed text and returns a category and confidence. The system stores the output and sends uncertain cases for agent confirmation.

4.4.3 Functional Requirements

ID	Requirement
FR-CLS-01	The system shall assign one approved category to each valid ticket.
FR-CLS-02	The system shall store predicted label, confidence where supported, model version, and inference timestamp.
FR-CLS-03	The system shall allow an agent to accept or correct the category.
FR-CLS-04	The system shall flag predictions below the configured confidence threshold for manual review.
FR-CLS-05	The system shall evaluate category performance separately for English, Sinhala, Tamil, Singlish, Tanglish, and available code-mixed sets.

4.5 Urgency and Sentiment Prediction

4.5.1 Description and Priority

Predicts operational priority and customer sentiment using approved labels and separately evaluated models or a documented multitask model. **Priority:** High

4.5.2 Stimulus/Response Sequences

The system analyses ticket text and available metadata, returns priority and sentiment results, and displays them to the agent. The agent may correct either result.

4.5.3 Functional Requirements

ID	Requirement
FR-ANA-01	The system shall assign a configured priority such as Low, Medium, High, or Critical.
FR-ANA-02	The system shall assign an approved sentiment label such as Negative, Neutral, or Positive.
FR-ANA-03	The system shall store task label, confidence where supported, model version, and correction history.
FR-ANA-04	The system shall permit deterministic escalation rules for explicitly defined critical terms or conditions.
FR-ANA-05	Sentiment alone shall not trigger a critical escalation.
FR-ANA-06	The system shall not train urgency or sentiment models until their label source and annotation method are documented.

4.6 Image Evidence Processing

4.6.1 Description and Priority

Extracts supplementary text or visual evidence from an optional image. **Priority:** Medium

4.6.2 Stimulus/Response Sequences

The system validates the image, performs OCR or approved vision processing, stores extracted evidence separately, and reports success, confidence, or failure. Text-based handling continues if processing fails.

4.6.3 Functional Requirements

ID	Requirement
FR-IMG-01	The system shall validate image format, size, readability, and safety before analysis.
FR-IMG-02	The system shall extract visible text using OCR when useful to the ticket.
FR-IMG-03	The system shall associate extracted evidence and processing status with the original image and ticket.
FR-IMG-04	Image-derived evidence shall not silently override explicit customer text or agent judgement.
FR-IMG-05	Failure of optional image processing shall not prevent category, priority, sentiment, or response processing from the text.

4.7 Ticket Routing and Lifecycle

4.7.1 Description and Priority

Creates an actionable ticket, assigns it to a queue, and tracks it until resolution. **Priority:** High

4.7.2 Stimulus/Response Sequences

After analysis, the system selects a queue using category and priority. An agent reviews, reassigns, escalates, adds notes, responds, resolves, or reopens the ticket. Each action is recorded.

4.7.3 Functional Requirements

ID	Requirement
FR-LIF-01	The system shall map approved categories to configured teams or queues.
FR-LIF-02	The system shall support manual assignment, reassignment, and escalation by an authorised user.
FR-LIF-03	The system shall support at least New, In Review, Assigned, Escalated, Resolved, and Closed statuses.
FR-LIF-04	The system shall retain timestamped status, assignment, correction, note, and response history.
FR-LIF-05	Higher-priority tickets shall appear before lower-priority tickets when other queue rules are equal.

4.8 Context-Aware Response Generation

4.8.1 Description and Priority

Generates an agent-editable draft from ticket text, predictions, optional image evidence, and approved knowledge. **Priority:** High

4.8.2 Stimulus/Response Sequences

An agent requests or opens a draft. The system generates a response, identifies it as machine-generated, and allows editing, regeneration, approval, or rejection before it becomes the final response.

4.8.3 Functional Requirements

ID	Requirement
FR-RSP-01	The system shall generate an editable response draft using available ticket context.
FR-RSP-02	The system shall support output in English, Sinhala, or Tamil when supported by the selected model.
FR-RSP-03	The system shall clearly mark generated content as a draft.
FR-RSP-04	The system shall require authorised human approval before a generated response is sent or recorded as final.
FR-RSP-05	The generator shall not invent balances, transaction status, approvals, personal data, or actions absent from authorised context.
FR-RSP-06	The system shall store the final response and, where permitted, its relationship to the generated draft for evaluation.

4.9 Agent Dashboard, Feedback, and Evaluation

4.9.1 Description and Priority

Supports review, correction, search, project evaluation, and basic model traceability. **Priority:** Medium

4.9.2 Stimulus/Response Sequences

An agent filters tickets and corrects outputs. An administrator or evaluator selects a dataset/model version and views aggregate and per-language results.

4.9.3 Functional Requirements

ID	Requirement
FR-EVL-01	The system shall support search/filter by ticket ID, text, category, priority, sentiment, language form, status, and date.
FR-EVL-02	The system shall record authorised corrections without automatically adding them to training data.
FR-EVL-03	The system shall display accuracy, macro precision, macro recall, macro F1-score, class-wise results, and confusion matrices for category classification.
FR-EVL-04	The system shall report suitable task metrics for priority and sentiment and compare results across supported language forms.
FR-EVL-05	The system shall record dataset version, split, preprocessing configuration, model version, and evaluation date.
FR-EVL-06	Only authorised users shall change model versions, thresholds, or label mappings.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-PER-01	Under normal demonstration load, 95% of text-only tickets shall receive category, priority, and sentiment outputs within 5 seconds.
NFR-PER-02	The agent ticket list shall load within 3 seconds for project-scale data and typical filters.
NFR-PER-03	Image processing and response generation shall provide timeout or progress feedback when processing exceeds 10 seconds.
NFR-PER-04	A failed optional model service shall not corrupt or delete the stored ticket.
NFR-PER-05	Batch evaluation shall be repeatable using recorded dataset, split, preprocessing, and model versions.

5.2 Safety Requirements

ID	Requirement
NFR-SAF-01	The system shall not execute financial transactions or account changes.
NFR-SAF-02	Low-confidence, ambiguous, or critical tickets shall be highlighted for human review.
NFR-SAF-03	A generated response shall not state that an action was completed unless verified by authorised data.
NFR-SAF-04	Original customer text and relevant model outputs shall be retained for investigation and correction.

5.3 Security Requirements

ID	Requirement
NFR-SEC-01	Network communication shall use TLS where the system is deployed over a network.
NFR-SEC-02	The system shall enforce least-privilege, server-side authorisation.
NFR-SEC-03	The system shall validate and sanitise text, files, queries, and API input.
NFR-SEC-04	Attachments and sensitive ticket data shall be accessible only to authorised users.
NFR-SEC-05	Secrets and credentials shall not be stored in source code, logs, generated text, or client bundles.
NFR-SEC-06	Security-sensitive, model-configuration, and final-response actions shall be auditable.

5.4 Software Quality Attributes

ID	Requirement
NFR-QUA-01	Usability: a trained agent shall complete ticket review and response without technical documentation.
NFR-QUA-02	Maintainability: classification, priority, sentiment, image, and generation components shall use replaceable documented interfaces.
NFR-QUA-03	Testability: each functional requirement shall be traceable to at least one test case.
NFR-QUA-04	Reliability: partial service failure shall preserve ticket data and expose a clear processing status.
NFR-QUA-05	Portability: the backend should support deployment on a standard Linux environment.
NFR-QUA-06	Explainability: authorised users shall see labels, confidence where available, model version, and correction status.
NFR-QUA-07	Fairness: evaluation shall report language-specific performance and identify material gaps.

5.5 Business Rules

1. Only an authorised agent may approve a final response or mark a ticket resolved.
2. Critical tickets and low-confidence predictions require explicit agent review.
3. Model corrections become candidate feedback, not automatically approved training data.
4. Category, priority, sentiment, and generated text shall be presented as model outputs, not guaranteed facts.
5. Model or threshold changes require an authorised user and a recorded version.

6. Other Requirements

6.1 Data Requirements

The system shall maintain tickets, original and processed text, optional attachments, predictions, confidence values, model versions, response drafts, final responses, assignments, statuses, corrections, evaluation results, and audit events. Dataset records shall retain source ID, language form, translation/transliteration lineage, label, and split. Derived versions of the same Banking77 source record shall remain in one split.

6.2 Internationalisation Requirements

All stored and transmitted text shall use UTF-8. The UI shall correctly display English, Sinhala, Tamil, Latin-script Singlish and Tanglish, and mixed-script content. The system shall not assume that one ticket contains only one language.

6.3 Privacy and Retention Requirements

The prototype shall collect only information needed for ticket processing and evaluation. Personal and financial information shall be minimised, access-controlled, and excluded from public datasets or reports. Research/training data shall be de-identified where practicable. Retention periods for tickets, images, prompts, outputs, and feedback shall be configurable or documented.

6.4 Model Training and Evaluation Requirements

- Category experiments shall use a held-out test set and report aggregate, class-wise, and per-language metrics.
- The project shall compare English performance with Sinhala, Tamil, Singlish, and Tanglish performance under the same split lineage.
- Translation quality and label preservation shall be checked through a documented validation sample.
- Naturally written code-mixed data, when available, shall be evaluated separately from translated or synthetically mixed data.
- Priority and sentiment datasets, labels, baselines, metrics, and acceptance thresholds shall be documented independently of Banking77.
- LLM responses shall be evaluated for relevance, language appropriateness, factual grounding, safety,

and usefulness to agents.

A. Glossary

Term	Definition
Banking77	Intent-classification dataset containing 77 online-banking customer-query categories.
Code-mixed text	One message containing elements from more than one supported language or script.
Singlish	Sinhala represented partly or fully using Latin characters.
Tanglish	Tamil represented partly or fully using Latin characters.
NLP	Natural Language Processing.
OCR	Optical Character Recognition.
LLM	Large Language Model.
Macro F1-score	Unweighted mean of per-class F1-scores.
Translation leakage	Inflated evaluation caused by placing translated versions of one source record in different data splits.
Confidence threshold	Configured value below which a prediction is routed for manual review.
TBD	To Be Determined.

B. Analysis Models

The final SRS package should include or reference the following diagrams when available:

- system context/use-case diagram;
- end-to-end ticket-processing data-flow diagram;
- multilingual model-training and inference architecture;
- ticket state-transition diagram;
- data or entity-relationship model;
- deployment diagram for the prototype.

C. To Be Determined List

1. Final university, department, author names, and approval date.
2. Whether the prototype uses all 77 Banking77 categories or an approved subset/mapping.
3. Final multilingual encoder and preprocessing/transliteration approach.
4. Source and annotation procedure for urgency and sentiment labels.
5. Final OCR/vision component and LLM response-generation component.
6. Confidence thresholds and quantitative acceptance targets for each task.
7. Maximum image size, retention period, and deployment hardware.
8. Availability and size of naturally written code-mixed evaluation data.